

ARAP · AIRNAV RISK ANALYSIS PLATFORM

User Guide

How to plan, run, review and approve safety risk assessments
with the twelve methods of the AirNav Risk Analysis Manual

Software version 0.1 · Guide edition 1 · September 2026

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ARAP User Guide

AirNav Risk Analysis Platform — version 0.1

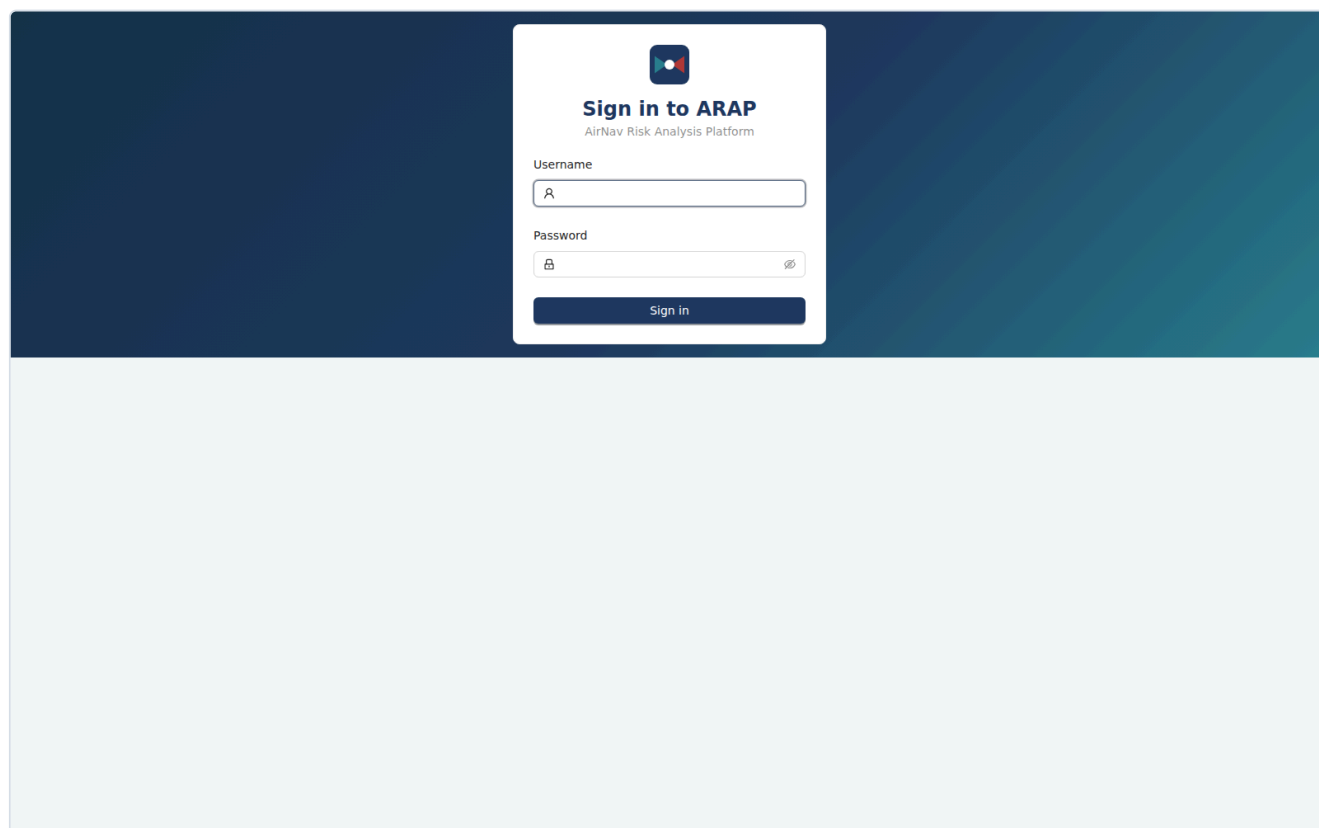
ARAP is the web application that implements the *AirNav Risk Analysis Manual*. It lets AirNav staff plan, run, review and approve safety risk assessments with twelve methods, keeps every hazard in one hazard log, and produces the Safety Assessment Report.

This guide explains how to use the application. For *why* and *when* to use each method, see the Manual ([docs/AirNav_Risk_Analysis_Manual.pdf](#)). For the formal requirements, see the SRS ([docs/AirNav_Risk_Analysis_Software_Requirements.pdf](#)).

1. Getting started

1.1 Signing in

Open ARAP in a browser (Chrome, Edge, Firefox or Safari) at the address your administrator gives you — for a local installation this is `http://localhost:8080` . Enter your username and password.



Sign-in page

1.2 Roles

What you can do depends on your role:

Role	Can do
Viewer	Read everything in their scope; update actions they own.
Assessor	Create projects, assessments and method studies; edit worksheets and diagrams; send hazards to the hazard log; raise actions; submit assessments for review.
Reviewer	Everything an assessor can, plus return, endorse or reject assessments, and read the audit trail.
Authority	Accept (or reject) residual risk, but only for the risk regions assigned to them (for example <i>Tolerable (lower)</i>).
Admin	Manage users, roles and the risk classification scheme. Admin is not a substitute for the proper acceptance authority in normal use.

1.3 Demo accounts

When ARAP is installed with demo data (`ARAP_SEED_DEMO=true`), these accounts exist, all with password `demo1234` :

Username	Role	May accept
assessor	Assessor	—
reviewer	Reviewer	—
director	Authority	Tolerable (lower), Acceptable
accexec	Authority (Accountable Executive)	Tolerable (upper), Tolerable (lower), Acceptable
viewer	Viewer	—

The administrator account is `admin` ; its password is set at installation. **Disable the demo data and change all passwords before real use.**

1.4 Screen layout and language

- The **left menu** gives access to the Dashboard, Projects & assessments, Hazard log, Actions, Risk scheme and User guide (plus Users and Audit trail for administrators and reviewers).
- The **EN / ID** switch at the top right changes the interface language between English and Bahasa Indonesia. Your choice is remembered on that computer. Text you type into worksheets is stored as entered.
- Your name and role are shown top right; click them to sign out.



3. Projects and assessments

3.1 Projects

A **project** represents one change: a new system, a procedure, an airspace change, an organisational change or a temporary change. Go to **Projects & assessments** → **New project** and enter a code (e.g. ADSB-2027), a title, the type of change, the units affected and the sponsor.

ARAP
AirNav Risk Analysis Platform

EN ID Ayu Assessor assessor

Projects & assessments

A project is a change (system, procedure, airspace, organisational). Each has one or more safety assessments.

[+ New project](#)

Projects

Code	Title	Type	Units	Assessments
APP-RELOC	Relocation of Jakarta approach unit to new ATC centre	organisational	Jakarta APP	1
DEMO-01	Manual worked examples	system	Jakarta ACC; Soekarno-Hatta TWR	1

All assessments

Project	Assessment	Version	Status	Updated
APP-RELOC	Transition safety assessment	1	in review	2026-09-28 10:22
DEMO-01	Worked examples — all twelve methods	1	draft	2026-09-28 10:22

< 1 >

Projects

3.2 Assessments

Open the project and choose **New assessment**. Describe the **scope and system boundary**, the **operational environment** and your **assumptions** (Manual §2.3). The assessment records which version of the risk scheme it uses.

The assessment page brings everything together:

Worked examples — all twelve methods draft v1

Submit for review | Report (.docx)

Scope, environment and assumptions Edit

Scope: Demonstration of every ARAP method module using the Manual's examples.
 Environment: Illustrative values only.
 Assumptions: All numbers are illustrative (Manual, How to use).
 Risk scheme: version 1

Method studies (12) Add study

Suggested for a system change (Manual §4.1) **HAZID** **FHA** **FMEA / FMECA** **FTA** **Bowtie analysis**

Method	Study	Status	Updated
Bowtie analysis	Runway incursion bowtie (Manual §6)	open	2026-09-28 10:22
HAZID	HAZID — approach unit relocation (Manual §7.4)	open	2026-09-28 10:22
HAZOP	HAZOP — estimate message coordination (Manual §8.3)	open	2026-09-28 10:22
JHA	JHA — obstruction light on radar tower (Manual §9.3)	open	2026-09-28 10:22
FMEA / FMECA	FMEA — sector VHF air-ground (Manual §10.3)	open	2026-09-28 10:22
LOPA	LOPA — level bust (Manual §11.3)	open	2026-09-28 10:22
FHA	FHA — ADS-B en-route surveillance (Manual §12.3)	open	2026-09-28 10:22
STPA	STPA — descent clearances (Manual §13.2)	open	2026-09-28 10:22
FTA	FTA — loss of surveillance picture (Manual §14.3)	open	2026-09-28 10:22
Fatigue modeling	Fatigue — night shift with/without nap (Manual §15.4)	open	2026-09-28 10:22
FRAM	FRAM — descent clearances in a busy arrival sector (Manual §16)	open	2026-09-28 10:22
Bayesian belief network	BBN — coordination error and loss of separation (Manual §17.3)	open	2026-09-28 10:22

Hazards (5) Open hazard log →

Ref	Hazard	Initial	Residual	Controls
HZ-0001	Unauthorised entry onto active runway	3A	2A	2
HZ-0002	Level bust leading to potential mid-air collision	3A	3A	2
HZ-0003	Total loss of surveillance picture at ACC sector	3B	3B	1
HZ-0004	ATCO fatigue at end of night shift	4C	3C	1
HZ-0005	New site outside radio line of sight of low-level sector	3B	—	0

Risk acceptance

Worst current/residual region: **Tolerable (upper)**
 Acceptance authority: **Accountable Executive**

Actions (5) All actions →

- ACT-0001 Install diverse, physically separate surveillance data path
CNS engineering - due 2026-12-27 - open
- ACT-0002 Implement SFL mismatch alert in ATM system
ATIS systems - due 2027-02-25 - open
- ACT-0003 Collect KSS survey for two roster cycles
FRMS specialist - due 2026-09-18 - open
- ACT-0004 Audit stop-bar U/S procedure compliance
Aerodrome ops - due 2026-09-25 - in progress
- ACT-0005 Radio coverage study for new site
CNS engineering - due 2026-10-28 - open

Review and approval history

Not yet submitted.

Assessment page

- **Method studies** — the analyses carried out for this assessment. ARAP suggests methods for the type of change (Manual §4.1); the suggestion is only advice.
- **Hazards** — hazard-log entries that belong to this assessment, with initial and residual risk.
- **Risk acceptance** — the worst residual risk region and the authority that must accept it.
- **Actions** and the **review and approval history**.
- Workflow buttons at the top right (see section 15) and **Report (.docx)**.

3.3 Adding a method study

On the assessment page choose **Add study**, pick the method (suggested methods are marked ★) and give the study a title, e.g. *HAZOP of cut-over plan*. The study opens straight away.

4. Working in a method study

Every study page has the same header:

- The **method** tag and the **study title** (editable).
- A link back to the assessment, the Manual chapter for the method and the template version.
- The number of **participants** recorded.
- **Save**. Changes are not saved automatically: the button reads *Save* while there are unsaved changes and *Saved* afterwards. If you try to leave the page with unsaved changes ARAP asks you to confirm.

When the assessment is endorsed, accepted or closed, or if your role cannot edit, the study opens **read-only** and a blue banner says so.

4.1 Sending results to the hazard log

Every method has a **Send to hazard log** button. It creates a hazard-log entry for each selected worksheet row (or for the bowtie as a whole, or for each STPA hazard), carrying over the causes, consequences, controls and any severity and likelihood. Sending the same row again **updates** the existing hazard rather than creating a duplicate. A message lists the hazard references created, e.g. HZ-0007, HZ-0008 . Rate the risk and complete the controls in the [hazard log](#).

5. Worksheet methods

HAZID, HAZOP, JHA, FMEA/FMECA and FHA use a spreadsheet-style worksheet.

Editing the worksheet

- **Double-click** a cell (or press Enter) to edit it; press Enter or click elsewhere to finish.
- Columns with fixed values — severity, likelihood, guideword, control type, failure type — show a drop-down list.
- **Add row** appends a row with the next ID (HZ-06 , HP-04 , ...).
- Tick the checkboxes on the left to select rows, then **Delete selected** or **Send ... to hazard log**.
- Click a column header to sort; use the filter icon to filter.
- The **Risk** column is computed from severity and likelihood using the current risk matrix.

The screenshot displays the ARAP AirNav Risk Analysis Platform interface. On the left is a dark sidebar with navigation links: Dashboard, Projects & assessments, Hazard log, Actions, Risk scheme, and User guide. The main area shows a HAZID worksheet titled 'HAZID — approach unit relocation (Manual §7.4)'. At the top right of the main area, it says 'EN ID' and 'Ayu Assessor assessor'. Below the title, there's a status bar with 'Worked examples — all twelve methods · Manual chapter 7 · template v1.0', '2 participants', and a 'Saved' button. The worksheet table has columns: ID, Guideword, Hazard, Cause(s), Consequence, Existing controls, S, L, Risk, and Actions. There are five rows of data. Rows HZ-01 and HZ-05 are selected, indicated by checkboxes in the first column. Below the table is a 'Guideword coverage' panel with a summary bar showing counts for various guidewords: People: 1, Procedures: 0, Equipment / systems: 1, Airspace and aerodrome: 0, Traffic: 0, Environment: 1, Interfaces: 1, Transition: 1, Organisation: 0. Below this bar is a dropdown menu with the text 'Record "no hazard identified" for a guideword'.

ID	Guideword	Hazard	Cause(s)	Consequence	Existing controls	S	L	Risk	Actions
HZ-01	Transition	Loss of surveillance display at cut-over	Data feed not re-routed; configuration error	Loss of separation during cut-over	Cut-over at low traffic; old site on hot standby	C	3	3C	HAZOP of cut rollback criteria
HZ-02	People	ATCOs unfamiliar with new CWP HMI	Short training window	Delayed or erroneous inputs	Simulator training	D	4	4D	Minimum hours before live ops
HZ-03	Interfaces	Wrong telephone coordination lines to TWR	VCCS mapping error	Late coordination of missed approach	Test plan	C	3	3C	End-to-end line each TWR
HZ-04	Equipment / systems	Single power feed to new building	Design	Total loss of service	UPS 30 min	B	2	2B	Second feed of power
HZ-05	Environment	New site outside radio line of sight of low-level sector	Topography	Loss of VHF contact below FL050	None yet	B	3	3B	Coverage study site

Guideword coverage

People: 1 Procedures: 0 Equipment / systems: 1 Airspace and aerodrome: 0 Traffic: 0 Environment: 1 Interfaces: 1 Transition: 1 Organisation: 0

Record "no hazard identified" for a guideword

HAZID worksheet after sending two rows to the hazard log

5.1 HAZID (Manual §7)

Work through the guidewords (People, Procedures, Equipment/systems, Airspace and aerodrome, Traffic, Environment, Interfaces, Transition, Organisation). The **Guideword coverage** panel below the worksheet shows how many rows each guideword has. When a guideword produces nothing, record that explicitly with **Record "no hazard identified"**. Use the *Needs method* column to flag rows that need a detailed method.

5.2 HAZOP (Manual §8)

1. In **Nodes and design intent**, add a node per procedure step or message exchange (+ **Node**), choose its type (*procedural* or *data-flow*) and write the design intent.
2. Press **Deviations** on a node. Choose the parameters and guidewords; ARAP adds one worksheet row per combination (e.g. *Message — No / Not*).

- For each row either tick **N/A** (not meaningful) or record causes, consequences, safeguards, S, L and a recommendation.
- The **Progress** column counts rows analysed out of rows generated.

The screenshot shows the ARAP (AirNav Risk Analysis Platform) interface. On the left is a dark sidebar with navigation links: Dashboard, Projects & assessments, Hazard log, Actions, Risk scheme, and User guide. The main workspace is titled 'HAZOP — estimate message coordination (Manual §8.3)'. It includes a 'Nodes and design intent' panel with a table showing nodes (N3), types (data-flow), and design intents. Below this is a main table with columns for ID, Node, Parameter, Guideword, Deviation, Possible causes, Consequences, Safeguards, S, L, and Progress. The table contains three rows of data (HP-01, HP-02, HP-03) related to message coordination. The interface also shows user information (Ayu Assessor) and a 'Saved' status.

ID	Node	Parameter	Guideword	Deviation	Possible causes	Consequences	Safeguards	S	L	Progress
HP-01	N3	Message	No / Not	Estimate not received	Link failure; flight plan not activated; address error	Unknown traffic enters sector	Link-status alarm; verbal coordination backup	C	3	3/3 analysed
HP-02	N3	Content	Other than	Coordinated level differs from actual	Level changed after EST; no revision	Level conflict downstream	Mode C/S check; STCA	B	3	
HP-03	N3	Timing	Late	Estimate inside agreed time	Short-notice route change	Workload	LoA fallback; verbal	D	4	

HAZOP worksheet with node panel

5.3 JHA (Manual §9)

Fill in the job header (job, location, permits), then one row per job step with its hazards, controls, control type and the person responsible. ARAP warns when a step relies only on **Administrative** controls or **PPE**, the lowest levels of the hierarchy of controls.

5.4 FMEA / FMECA (Manual §10)

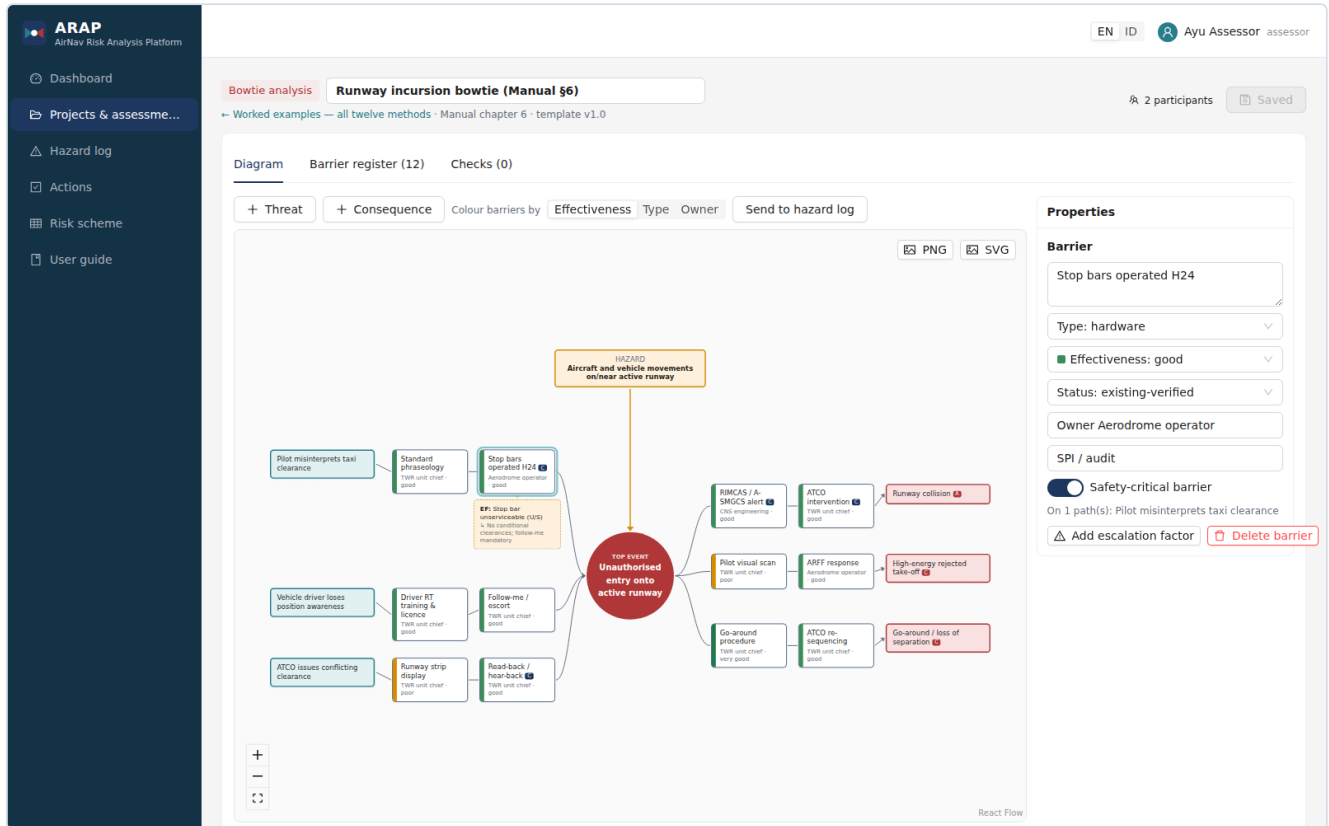
One row per failure mode: item, failure mode, cause, end effect on the ATS, detection, compensation, and **S, O, D** ratings (1–10). ARAP computes the **RPN** ($S \times O \times D$), colours high values and sorts by RPN. It also lists high-severity modes ($S \geq 8$) so they are not missed because of a low RPN. For FMECA, add the failure rate λ_p , mode ratio α , conditional probability β and operating time t .

5.5 FHA (Manual §12)

One row per failure condition: the function, the failure type (total loss, partial loss, erroneous detected/undetected, unintended, delayed), the condition, the operational effect and the **severity**. The **Safety objective** column is computed from the risk scheme: the maximum frequency per operating hour that keeps the failure condition at or better than *Tolerable (lower)* (Manual Table 12.1). Record the derived safety requirements in the last column.

6. Bowtie analysis

The bowtie editor draws the diagram automatically: threats on the left, the top event in the centre, consequences on the right, barriers in order on each line and escalation factors below the barrier they weaken.



Bowtie editor with a barrier selected

Building the bowtie

1. Click the **HAZARD** box and the **TOP EVENT** circle to edit their text. Phrase the top event as the moment control is lost, *before* any damage.
2. **+ Threat** and **+ Consequence** add lines. Click one to edit its text (and a consequence's severity).
3. With a threat or consequence selected, the **Properties** panel lists its barriers in order: - **New barrier on this path** adds a barrier; - **Add an existing (shared) barrier** puts a barrier that already exists on this line too — a shared barrier shows **xn**; - the arrows reorder barriers and the bin removes a barrier from this line only.
4. Click a **barrier** to set its type, effectiveness, status (existing-verified, existing-unverified, planned), owner, SPI/audit reference and whether it is **safety-critical** (C badge).
5. With a barrier selected, **Add escalation factor** attaches a condition that degrades it. Select the escalation factor to write its escalation-factor barriers, one per line.

Viewing

- **Colour barriers by** effectiveness (default), type or owner.
- Zoom with the mouse wheel or the **+** / **–** controls; drag the background to pan; the frame icon fits the diagram.
- **PNG** and **SVG** download the full diagram as an image.
- **Barrier register** lists every barrier with its attributes (Manual §6.3).

- **Checks** lists issues: a top event phrased like a consequence, lines with fewer than two barriers, barriers without an owner.

ARAP

AirNav Risk Analysis Platform

Dashboard

Projects & assessments

Hazard log

Actions

Risk scheme

User guide

EN ID

Ayu Assessor assessor

Bowtie analysis

Runway incursion bowtie (Manual §6)

2 participants

Saved

Diagram

Barrier register (12)

Checks (0)

Barrier	Side	Type	Owner	Effectiveness	Critical	Status	Paths	SPI / audit
Standard phraseology	prevention	human	TWR unit chief	good		existing-verified	Pilot misinterprets taxi clearance	
Stop bars operated H24	prevention	hardware	Aerodrome operator	good	Yes	existing-verified	Pilot misinterprets taxi clearance	
Driver RT training & licence	prevention	human	TWR unit chief	good		existing-verified	Vehicle driver loses position awareness	
Follow-me / escort	prevention	human	TWR unit chief	good		existing-verified	Vehicle driver loses position awareness	
Runway strip display	prevention	human-hardware	TWR unit chief	poor		existing-verified	ATCO issues conflicting clearance	
Read-back / hear-back	prevention	human	TWR unit chief	good	Yes	existing-verified	ATCO issues conflicting clearance	
RIMCAS / A-SMGCS alert	recovery	hardware	CNS engineering	good	Yes	existing-verified	Runway collision	
ATCO intervention	recovery	human	TWR unit chief	good	Yes	existing-verified	Runway collision	
Pilot visual scan	recovery	human	TWR unit chief	poor		existing-verified	High-energy rejected take-off	
ARFF response	recovery	human-hardware	Aerodrome operator	good		existing-verified	High-energy rejected take-off	
Go-around procedure	recovery	human	TWR unit chief	very good		existing-verified	Go-around / loss of separation	
ATCO re-sequencing	recovery	human	TWR unit chief	good		existing-verified	Go-around / loss of separation	

Barrier register

7. LOPA

The screenshot displays the LOPA calculator interface. On the left is a sidebar with 'Risk scheme' and 'User guide'. The main area is divided into several sections:

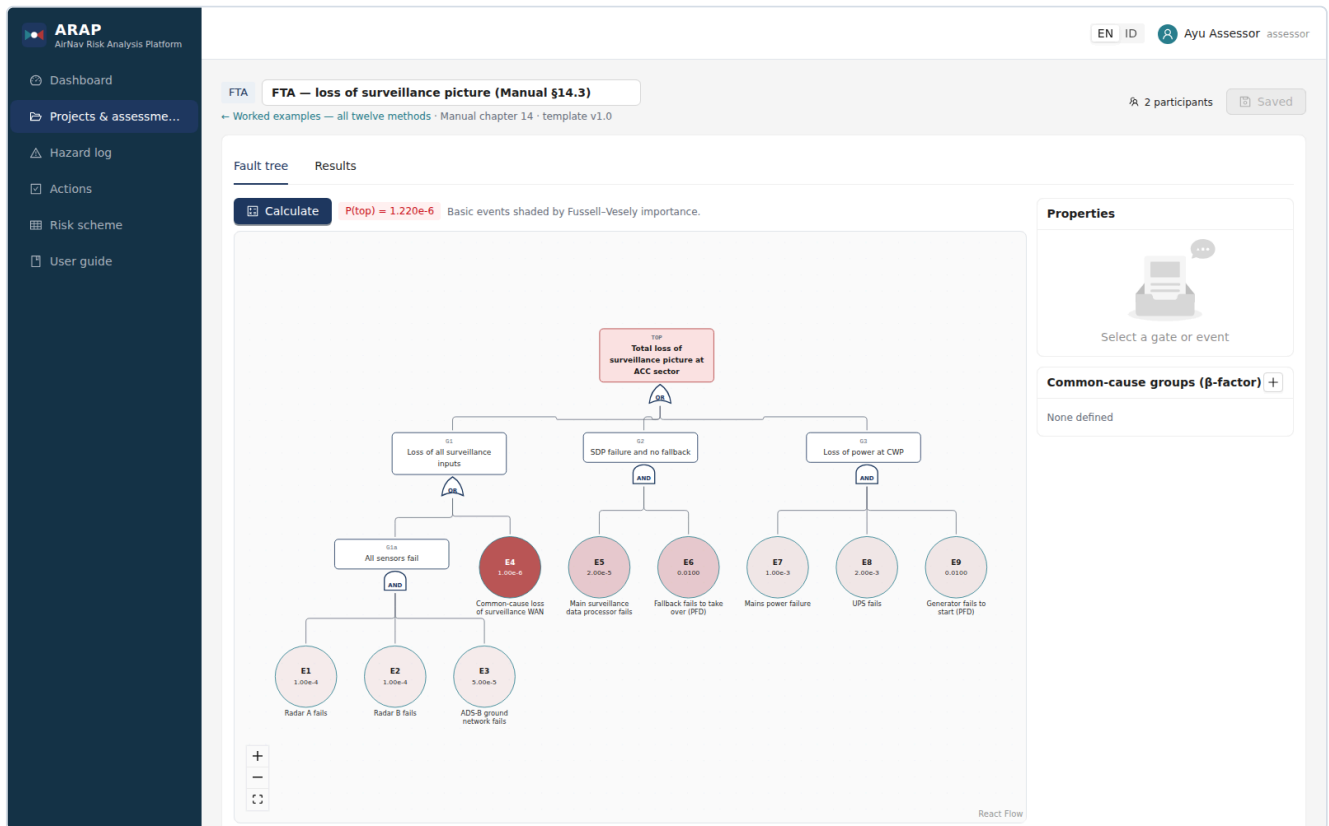
- Scenario Level:** 'bust leading to a potential mid-air collision'. Below this are fields for 'Initiating event Level bust in sector X', 'f(IE) 0.5', 'Unit per year', 'Data source Illustrative', and 'Tolerable target 0.00001'.
- Enabling conditions / conditional modifiers:** A table with 'Condition' and 'Probability'. One condition is 'Conflicting traffic present' with a probability of '0.05'.
- Safeguards — credited only if all four IPL criteria are met:** A table with columns: 'Safeguard', 'PFD', 'indep.', 'effec.', 'depen.', 'audit.', 'Shared dependencies', and 'Justi'. Safeguards listed include 'ATCO monitoring and intervention', 'STCA alert and ATCO resolution', 'ACAS II RA followed', 'SFL mismatch alert (proposed)', and 'Recurrent RT training'.
- Results:**
 - Mitigated frequency: **2.50e-5** per year
 - Target: 1.00e-5 per year
 - Verdict: **Gap x2.5**
 - Further reduction needed: x2.5 (0.40 orders of magnitude)
 - Credited IPLs: ATCO monitoring and intervention; STCA alert and ATCO resolution; ACAS II RA followed
- Warnings:**
 - Not credited: SFL mismatch alert (proposed) Missing: dependable
 - Not credited: Recurrent RT training Missing: independent, effective
- Justification Note:** IPLs ATCO monitoring and intervention, STCA alert and ATCO resolution share dependency 'surveillance' — independence must be justified
- Bar Chart:** A log-scale chart showing frequency (per year) for each layer: Initiating event (1e+0), Conflicting traffic present (5.0e-1), ATCO monitoring and intervention (2.5e-2), STCA alert and ATCO resolution (2.5e-3), and ACAS II RA followed (2.5e-4).
- Footer:** Engine lopa-1.0.0. LOPA is an order-of-magnitude screening tool (Manual §11.4).

LOPA calculator

1. Describe the **scenario**, the **initiating event**, its frequency, unit and data source, and the **tolerable target** frequency.
2. Add **enabling conditions / conditional modifiers** with their probabilities.
3. Add **safeguards** with a probability of failure on demand (PFD). Tick each of the four IPL criteria — *independent, effective, dependable, auditable* — only if it is justified, and write the justification. A safeguard is credited **only if all four are ticked**.
4. Record shared dependencies (e.g. **surveillance**) — ARAP warns if two credited layers share one.
5. Press **Calculate**. The result shows the mitigated frequency, the verdict (*Target met* or *Gap x n*), the extra risk reduction needed, the safeguards not credited and why, and a log-scale chart of the frequency after each layer.

In the demo study the proposed *SFL mismatch alert* is not yet ticked *dependable*, so the result shows a gap of x2.5. Tick it and recalculate to see the target met (2.5×10^{-6} per year).

8. Fault tree analysis



Fault tree

Building the tree

- Click a gate and use **OR gate**, **AND gate**, **Basic event** or **Undeveloped** in the Properties panel to add inputs. **Reuse an existing node as input** links a node already in the tree (e.g. a shared support system).
- For a **vote** gate set k (k -out-of- n).
- For a **basic event** give either a probability p , or a failure rate λ with an exposure time t ($P = 1 - e^{-\lambda t}$), a repair time MTTR (repairable unavailability), or a test interval T (periodically tested standby, $\lambda T/2$).
- A **house event** is a switch set to TRUE or FALSE.
- Common-cause groups** apply the β -factor model to a set of basic events.

Calculating

Press **Calculate**. The top-event probability appears above the tree, and basic events are shaded by their Fussell-Vesely importance. The **Results** tab gives:

- the exact top-event probability (binary decision diagram), the rare-event approximation and the min-cut upper bound;
- single points of failure;
- minimal cut sets with probability and share;
- Birnbaum and Fussell-Vesely importance for every basic event.

ARAP
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Dashboard

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ENID

Ayu Assessorassessor

FTA — loss of surveillance picture (Manual §14.3)

Worked examples — all twelve methods · Manual chapter 14 · template v1.0

2 participants

Save

Fault treeResults

Top-event probability (exact, BDD)	1.2200e-6
Rare-event approximation	1.2200e-6
Min-cut upper bound	1.2200e-6
Single points of failure	E4
Engine	fta-1.1.0 · 31 BDD nodes

Minimal cut sets

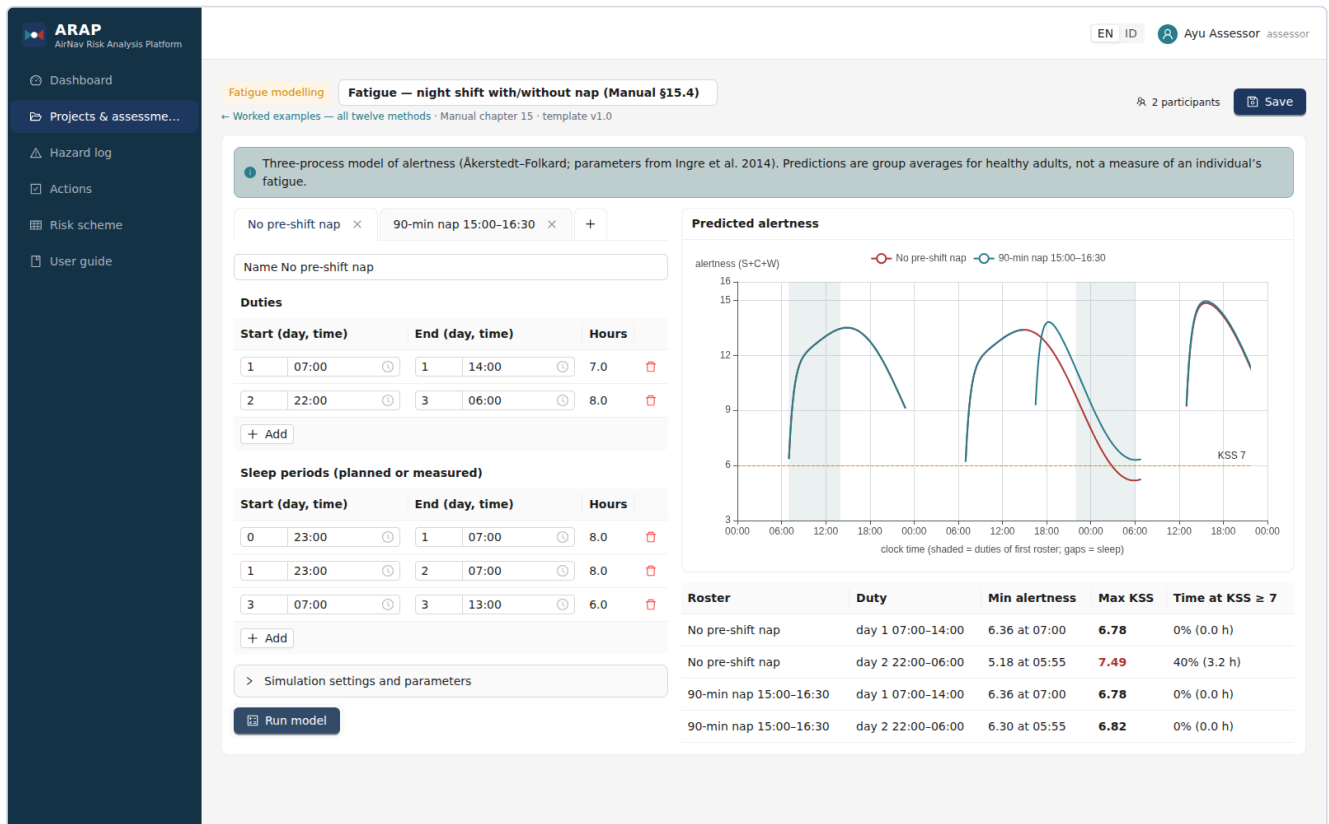
Cut set	Order	Probability	Share
{E4}	1	1.00e-6	82.0%
{E5, E6}	2	2.00e-7	16.4%
{E7, E8, E9}	3	2.00e-8	1.6%
{E1, E2, E3}	3	5.00e-13	0.0%

Importance measures

Event	Description	p	Birnbaum	Fussell-Vesely
E4	Common-cause loss of surveillance WAN	1.00e-6	1.00	0.820
E5	Main surveillance data processor fails	2.00e-5	1.00e-2	0.164
E6	Fallback fails to take over (PFD)	0.0100	2.00e-5	0.164
E7	Mains power failure	1.00e-3	2.00e-5	0.0164
E8	UPS fails	2.00e-3	1.00e-5	0.0164
E9	Generator fails to start (PFD)	0.0100	2.00e-6	0.0164
E1	Radar A fails	1.00e-4	5.00e-9	4.098e-7
E2	Radar B fails	1.00e-4	5.00e-9	4.098e-7
E3	ADS-B ground network fails	5.00e-5	1.00e-8	4.098e-7

FTA results

9. Fatigue modelling



Fatigue model

1. Each tab is a **roster variant** (e.g. *No pre-shift nap* and *90-min nap*). Use **+** to copy the first roster into a new variant.
2. Enter **duties** and **sleep periods** as day number and clock time. Day 1 starts at 00:00; a sleep starting at 23:00 the evening before day 1 is entered as day 0, 23:00.
3. **Simulation settings and parameters** sets the number of days, time step, KSS threshold and, if needed, the model parameters (leave blank for the defaults of the Manual, §15.2).
4. Press **Run model**. The chart shows predicted alertness for each variant (gaps are sleep; shaded bands are duties of the first variant; the dashed line is the KSS threshold). The table gives, per duty, the minimum alertness and when it occurs, the maximum KSS, and the time spent at or above the threshold.

Predictions are group averages for healthy adults. They are not a measure of an individual's fatigue and must be checked against operational data (Manual §15.5).

10. Bayesian belief networks

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Bayesian belief network **BBN — coordination error and loss of separation (Manu)** 2 participants Save

← Worked examples — all twelve methods · Manual chapter 17 · template v1.0

Network

+ Node Update beliefs

Drag from a node's bottom handle to another node to add a causal arc. Set evidence in the panel.

ATCO fatigue F
yes 39.01%
no 60.99%

Workload W
high 54.48%
normal 45.52%

Coordination error E
yes 100.00%
no 0.00%

STCA available S
yes 89.89%
no 10.91%

Loss of separation L
yes 100.00%
no 0.00%

Node L

Label Loss of separation

States yes, no

Evidence yes

Conditional probability table

Coordination error	STCA available	P(yes)	P(no)	Σ
yes	yes	0.050	0.950	1.0
yes	no	0.300	0.700	1.0
no	yes	0.000	1.000	1.0
no	no	0.000	1.000	1.0

Sensitivity of P(yes) Delete node

P(evidence) = 2.453e-4 · engine bbn-1.0.0

Bayesian network with evidence

Building the network

- **+ Node** adds a node; drag nodes to arrange them.
- To add a causal arc, drag from the **bottom handle** of the cause to the effect node. Double-click an arc to remove it. ARAP rejects cycles when calculating.
- Select a node to edit its label and **states** (comma-separated) and its **conditional probability table** — one row per combination of parent states. Each row must sum to 1; the Σ column turns red when it does not (for two-state nodes the second value fills in automatically).

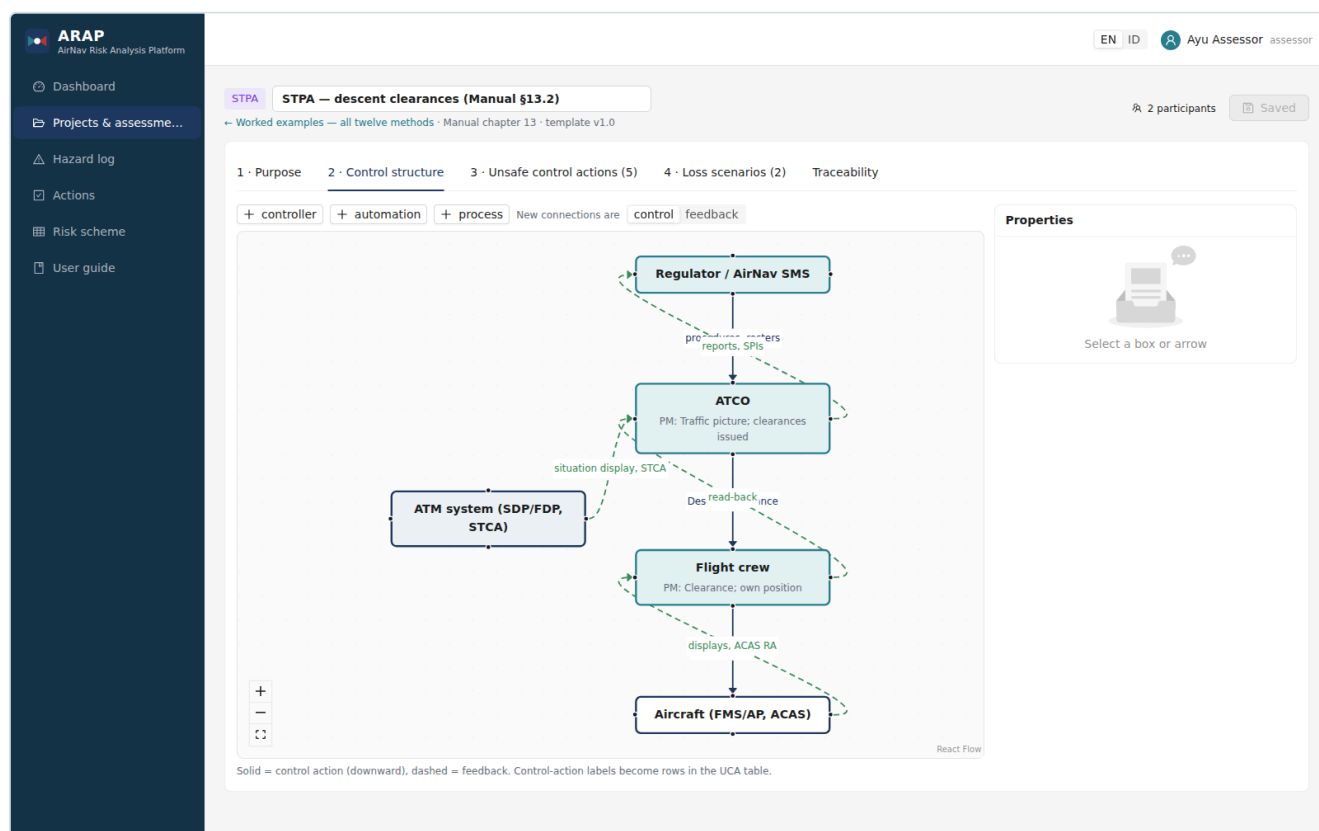
Using the network

- **Update beliefs** calculates every node's probabilities; the bars inside the nodes show them.
- Select a node and choose a state under **Evidence** to condition on it (the node turns amber). All other nodes update — this is how you ask diagnostic questions such as *given a loss of separation, how likely was fatigue?*
- **Sensitivity** shows how much each other node can move the selected node's probability (tornado chart).

11. STPA

The STPA study follows the four steps of the STPA Handbook, one tab each.

1. **Purpose** — losses, system-level hazards (linked to losses) and system-level constraints (linked to hazards).
2. **Control structure** — add controllers, automation and controlled processes; drag from a box's bottom/right handle to another box to add an arrow. Choose whether new arrows are **control** actions (solid) or **feedback** (dashed). Click a box to write its *process model*; click an arrow to rename it.



STPA control structure

3. **Unsafe control actions** — for every control action, four columns (*not providing, providing, too early/too late/wrong order, stopped too soon/applied too long*). Add UCAs with **+ UCA**; each needs a context ("when ...") and at least one hazard, otherwise ARAP warns.

ARAP

AirNav Risk Analysis Platform

Dashboard
Projects & assessments
Hazard log
Actions
Risk scheme
User guide

EN ID

Ayu Assessor assessor

STPA

STPA — descent clearances (Manual §13.2)

Worked examples — all twelve methods · Manual chapter 13 · template v1.0
2 participants
Saved

1 · Purpose

2 · Control structure

3 · Unsafe control actions (5)

4 · Loss scenarios (2)

Traceability

Control action: procedures, rosters

Not providing causes hazard

Providing causes hazard

Too early / too late / wrong order

Stopped too soon / applied too long

Control action: Descent clearance

Not providing causes hazard

UCA-1
ATCO does not issue descent
when the aircraft must leave its level to avoid a conflict ahead
H-1
remove
+ UCA

Providing causes hazard

UCA-2
ATCO clears descent through an occupied level
when unseparated traffic is at that level
H-1
remove
UCA-3
ATCO clears descent below MVA
when vectoring over high terrain
H-2
remove
+ UCA

Too early / too late / wrong order

UCA-6
Descent issued before preceding aircraft vacated level
when relying on the label trend
H-1
remove
+ UCA

Stopped too soon / applied too long

+ UCA

Control action: Execute descent

Not providing causes hazard

Providing causes hazard

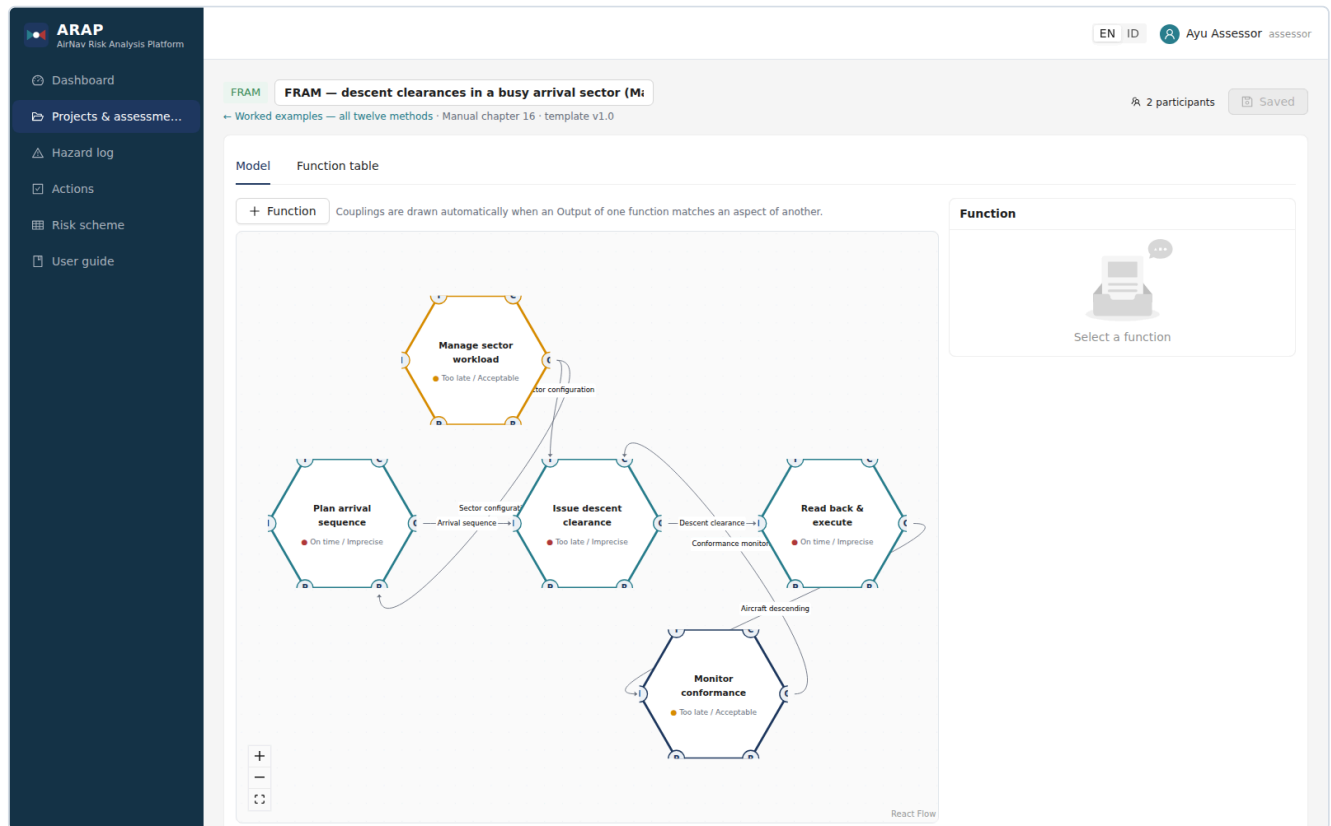
Too early / too late / wrong order

Stopped too soon / applied too long

UCA table

- Loss scenarios** — for each UCA, the scenario type, description and the resulting requirement or constraint. **Send hazards to hazard log** creates one hazard-log entry per STPA hazard.
- Traceability** — the chain loss → hazard → constraint/UCA → scenario → requirement, with broken chains flagged.

12. FRAM



FRAM model

- **+ Function** adds a function (hexagon). Drag to arrange.
- Select a function to set its name, type (human, technological, organisational) and the six **aspects** — Input, Output, Precondition, Resource, Control, Time. Type an entry and press Enter; existing outputs are offered as suggestions.
- **Couplings are drawn automatically** wherever an Output of one function has the same text as an aspect of another. Unmatched aspects are listed so you can fix typos or add missing functions.
- Record the **output variability** (timing: too early / on time / too late / not at all; precision: precise / acceptable / imprecise). The dot colour on each hexagon shows it.
- The **Function table** tab lists everything in one table.

13. The hazard log

ARAP
AirNav Risk Analysis Platform

Dashboard

Projects & assessments...

Hazard log

Actions

Risk scheme

User guide

EN ID Ayu Assessor assessor

+ New hazard

↓ CSV

Hazard log

Single register of hazards from every method study, with current and residual risk, controls and owners (Manual §18.2).

Search

Q

Risk region

6 of 6

Ref	Hazard	Unit	System	Owner	Initial	Residual	Controls	Review	Status
HZ-0001	Unauthorised entry onto active runway	Soekarno-Hatta TWR	Aerodrome	TWR unit chief	3A	2A	2	2026-10-18	open
HZ-0002	Level bust leading to potential mid-air collision	Jakarta ACC	ATM	ACC unit chief	3A	1A	2	2027-01-26	open
HZ-0003	Total loss of surveillance picture at ACC sector	Jakarta ACC	Surveillance	ACC unit chief	3B	3B	1	2026-11-12	open
HZ-0004	ATCO fatigue at end of night shift	Jakarta ACC	Roster	ACC unit chief	4C	3C	1	2026-09-23	open
HZ-0005	New site outside radio line of sight of low-level sector	Jakarta APP	VHF	ACC unit chief	3B	—	0	2026-11-27	open
HZ-0006	Loss of surveillance display at cut-over	Jakarta APP	Surveillance	Project manager	3C	2C	1	2026-12-27	open

< 1 >

Hazard log

The hazard log is the single register of hazards from all studies (Manual §18.2). Search, filter by risk region, sort, and export to **CSV**. Overdue review dates are shown in red.

Click a hazard to open it:

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Dashboard

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Hazard log

Actions

Risk scheme

User guide

Hazard log

Single register of hazards from every method study, with current

Search

Risk region

Ref	Hazard	Unit
HZ-0001	Unauthorised entry onto active runway	Soeka
HZ-0002	Level bust leading to potential mid-air collision	Jakarta
HZ-0003	Total loss of surveillance picture at ACC sector	Jakarta
HZ-0004	ATCO fatigue at end of night shift	Jakarta
HZ-0005	New site outside radio line of sight of low-level sector	Jakarta
HZ-0006	Loss of surveillance display at cut-over	Jakarta

×

HZ-0001 — Unauthorised entry onto active runway

Save

Details and risk

Controls (2)

Links

History

* Hazard

Unauthorised entry onto active runway

Causes / threats

Consequences

Unit

Soekarno-Hatta TWR

System

Aerodrome

Owner

TWR unit chief

Status

open

Review date

2026-10-18

Initial / current risk

3A

Residual risk

2A

clear

	Catastrophic Hazardous Major Minor Negligible						Catastrophic Hazardous Major Minor Negligible				
	A	B	C	D	E		A	B	C	D	E
Frequent 5	5A	5B	5C	5D	5E	Frequent 5	5A	5B	5C	5D	5E
Occasional 4	4A	4B	4C	4D	4E	Occasional 4	4A	4B	4C	4D	4E
Remote 3	3A	3B	3C	3D	3E	Remote 3	3A	3B	3C	3D	3E
Improbable 2	2A	2B	2C	2D	2E	Improbable 2	2A	2B	2C	2D	2E
Extremely improbable 1	1A	1B	1C	1D	1E	Extremely improbable 1	1A	1B	1C	1D	1E

Rationale and evidence for the ratings (required for residual risk)

Occurrence data 2023-2025; barrier audit results

Description / context

Hazard detail

- **Details and risk** — description, causes, consequences, unit, system, owner, status and review date. Rate **initial/current** and **residual** risk by clicking a cell in each matrix. A residual rating requires a **rationale** (the basis of the judgement).
- When you create a new hazard, ARAP checks for **possible duplicates** as you leave the title field.
- **Controls** — each with side (prevention/recovery), type, status, effectiveness, owner and a safety-critical flag. Only *existing-verified* controls may be credited in the current risk.
- **Links** — the source study, the assessment and the related actions.
- **History** — every change from the audit trail.

14. Actions

ARAP
AirNav Risk Analysis Platform

EN ID Ayu Assessor assessor

Actions
Risk-reduction actions with owners and due dates. Overdue actions are escalated on the dashboard (COM-12).

+ New action

Ref	Action	Hazard	Owner	Due	Status
ACT-0003	Collect KSS survey for two roster cycles	HZ-0004	FRMS specialist	2026-09-18	open
ACT-0004	Audit stop-bar U/S procedure compliance	HZ-0001	Aerodrome ops	2026-09-25	In progress
ACT-0005	Radio coverage study for new site	HZ-0005	CNS engineering	2026-10-28	open
ACT-0001	Install diverse, physically separate surveillance data path	HZ-0003	CNS engineering	2026-12-27	open
ACT-0002	Implement SFL mismatch alert in ATM system	HZ-0002	ATM systems	2027-02-25	open

< 1 >

Actions

Actions have an owner, a due date, a status (open, in progress, closed) and an optional link to a hazard. Overdue actions are flagged here and on the dashboard. **Closing an action requires closure evidence** (e.g. a report number). A Viewer can update an action when the action's owner field is their username.

15. Review, approval and locking

Status	Meaning	Next step and who does it
Draft	Being prepared; editable.	Submit for review — assessor
In review	Reviewer checking.	Endorse or Return to assessor (comment required) — reviewer
Endorsed	Reviewed; locked.	Accept residual risk — authority for the worst residual region; or Reject (comment required)
Accepted	Residual risk accepted; locked.	Close — reviewer or authority
Rejected	Not accepted.	Revise and Submit again
Closed / Superseded	Finished, or replaced by a newer version.	—

Rules ARAP enforces

- An assessment with an **intolerable** residual risk cannot be accepted.
- An assessment with **no hazards** cannot be accepted.
- The person accepting must hold authority for the **worst** residual risk region in the assessment (e.g. *Tolerable (upper)* needs the Accountable Executive).
- Endorsed, accepted and closed assessments are **locked**: studies and hazards become read-only. To change anything, use **New version**. This copies the studies into a new draft (version n+1) and marks the old version *superseded*.

The screenshot shows the ARAP interface for a 'Transition safety assessment' (version 1) which is in an 'accepted' state. The interface includes a sidebar with navigation options: Dashboard, Projects & assessments, Hazard log, Actions, Risk scheme, and User guide. The main content area displays the assessment details, including the scope, environment, and assumptions. It also shows a table of method studies and a table of hazards. The 'Risk acceptance' section indicates the worst current/residual region is 'Tolerable (lower)' and the acceptance authority is 'Director of Operations / Engineering'. The 'Actions' section shows no actions. The 'Review and approval history' section shows a sequence of events: endorsement by a reviewer and acceptance by the director.

Accepted assessment

Every transition is recorded in the **review and approval history** and the audit trail.

16. Reports and exports

What	Where	Format
Safety Assessment Report (Manual §18.3)	Assessment page → Report (.docx)	Word
Hazard log	Hazard log → CSV	CSV (opens in Excel)
Bowtie diagram	Bowtie editor → PNG / SVG	Image

The report includes scope, methods used, hazards and their risk ratings, study results (fault tree cut sets, LOPA verdict, fatigue indicators, Bayesian network probabilities), actions, the approval history, and the versions of the risk scheme and calculation engines used.

17. Administration

17.1 Risk scheme

Risk scheme shows the matrix, the tolerability regions with their required action and acceptance authority, and the likelihood levels with their quantitative bands.

The screenshot displays the ARAP (AirNav Risk Analysis Platform) interface. The left sidebar contains navigation links: Dashboard, Projects & assessments, Hazard log, Actions, Risk scheme (selected), and User guide. The main content area is titled 'Risk scheme' and includes a subtitle 'Active version 1. Changes create a new version and never alter locked assessments (ADM-01)'. The interface is divided into three main sections:

- Risk matrix:** A 5x5 grid of risk cells. The columns are labeled 'Catastrophic', 'Major', 'Minor', and 'Negligible'. The rows are labeled 'Frequent', 'Occasional', 'Remote', 'Improbable', and 'Extremely improbable'. The cells are color-coded: red for 'Intolerable' (5A, 5B, 5C), orange for 'Tolerable (upper)' (3A, 4A, 4B, 5A, 5B, 5C), yellow for 'Tolerable (lower)' (1A, 2A, 2B, 2C, 3D, 4D, 4E, 5E), and green for 'Acceptable' (1B, 1C, 1D, 1E, 2D, 2E, 3E, 4C, 4D, 4E, 5D, 5E). A tooltip for cell 5B indicates 'Intolerable'.
- Tolerability and acceptance authority:** A table with four columns: Region, Cells, Required action, and Authority.
- Likelihood levels:** A table with four columns: Level, Name, Nama, and Band (per operating hour).

Region	Cells	Required action	Authority
Intolerable	3A, 4A, 4B, 5A, 5B, 5C	Do not introduce or continue; redesign or add controls.	Cannot be accepted
Tolerable (upper)	2A, 3B, 4C, 5D	Additional controls; ALARP justification; monitoring plan.	Accountable Executive
Tolerable (lower)	1A, 2B, 2C, 3C, 3D, 4D, 4E, 5E	Mitigate where reasonably practicable; document ALARP.	Director of Operations / Engineering
Acceptable	1B, 1C, 1D, 1E, 2D, 2E, 3E	Accept; monitor.	Unit / branch manager

Level	Name	Nama	Band (per operating hour)
5	Frequent	Sering	1e-3 and above
4	Occasional	Kadang-kadang	1e-5 to 1e-3
3	Remote	Jarang	1e-7 to 1e-5
2	Improbable	Tidak mungkin	1e-9 to 1e-7
1	Extremely improbable	Sangat tidak mungkin	< 1e-9

Risk scheme

Administrators can reassign cells: choose a region under **Paint region**, click the cells, then **Save as new version**. ARAP checks that every cell belongs to exactly one region. Existing assessments keep the version they were created with.

17.2 Users

Users (administrators only) lists accounts. Click one to edit, or **New user** to add one. Set the role, unit and — for authorities — the risk regions they may accept. Deactivate rather than delete leavers.

17.3 Audit trail

Audit trail (reviewers and administrators) lists every create, change, approval and export with the user, time and changed fields. Filter by entity.

18. Troubleshooting and FAQ

I can't edit a study or hazard. Either your role is Viewer or Authority, or the assessment is endorsed/accepted/closed. Use **New version** on the assessment page (assessor or reviewer).

"A rationale is required for every risk judgement". Fill in *Rationale and evidence* in the hazard before saving a residual rating.

"Residual risk region ... requires ...". You do not hold authority for the worst residual risk in the assessment. Ask the named authority to accept it.

The worksheet did not keep my edit. Finish the edit (Enter or click another cell), then press **Save** in the header.

My session ended. Sessions last 8 hours by default; sign in again.

Numbers differ slightly from the Manual. The engines reproduce the Manual's worked examples within the tolerances in SRS §9. Rounding in the Manual text is to 2–3 significant figures.

19. What version 0.1 does not do yet

Version 0.1 covers the core workflow and all twelve methods. The following SRS items are planned for later releases:

- Single sign-on through Keycloak / SAML / OIDC with MFA. Version 0.1 uses local accounts with JWT tokens.
- Real-time co-editing of a diagram by several users, and the offline workshop pack.
- Imports and exports in BowTieXP XML, Open-PSA MEF, XMLBIF, GeNIe and FRAM Model Visualiser formats.
- File attachments (MinIO), e-mail notifications and reminders.
- Integrations with the occurrence reporting, rostering and maintenance systems.
- CAST (STPA for accident analysis), noisy-OR helpers for Bayesian networks, Monte Carlo uncertainty, and the full “braked” sleep-recovery form of the fatigue model.
- An Apache Superset / Metabase reporting database.